A Systematic Review of Antimicrobial Self-Medication in Nigeria

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ABSTRACT

Antimicrobial self-medication (ASM) has triggered a major global health crisis due to the associated risk of antimicrobial drug resistance. There is a high prevalence of ASM in low and middle-income countries including Nigeria, hence the review was undertaken to evaluate its impact and consequences. The study entails a systematic review of 18 research articles on antimicrobial self-medication in Nigeria using search engines such as Google Scholar, Medline, Hinari, PubMed, Ebscohost, Embase, and Scopus. A two-stage approach was used in selecting the articles using specifically defined inclusion and exclusion criteria. A schematic flow-chart was developed as recommended by the Prisma statement on conducting systematic reviews and meta-analysis and results were presented as proportions in tables and charts. The review showed that the prevalence of ASM in Nigeria was 51.4% with antimicrobials such as amoxicillin + clavulanic acid (66.7%), ampicillin + cloxacillin (55.6%), ampicillin (44.4%), metronidazole (27.8%), and ciprofloxacin (27.8%) topping the list as the most frequently misused antimicrobials in Nigeria. The burden of antimicrobial self-medication is high and it may have a grave impact on healthcare delivery both in Nigeria and globally. The adoption of a multisectoral approach as a public health intervention strategy will greatly help in mitigating the emerging risk of ASM in the country.

Keywords: Antimicrobials, Self-medication, Drug resistance, Multisectoral approach.

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1. INTRODUCTION

Antimicrobial self-medication (ASM) has become a global phenomenon. Studies have shown that the global prevalence of antimicrobial self-medication ranges from 4-45%[1,2,3]. However, data from low-and-middle-income countries (LMICs) shows a prevalence of 38.8%.[2] Self-medication is defined as the use of medicines for the treatment of self-diagnosed illnesses or symptoms of diseases[2,4,5]. Studies have shown that ASM is one of the major risk factors for the emergence of antimicrobial drug resistance[2,5,6]. Antimicrobials are chemical substances produced naturally from microbial components or synthesized as chemicals. At low concentrations, these chemical agents can inhibit the growth of microorganisms or kill microorganisms. They, however, undergo transmissible loss of efficacy following repeated use over time [1,7,8]. Once an antimicrobial is administered and is not able to effectively inhibit the growth of an organism, antimicrobial resistance is said to have occurred[1].

Several factors have been identified as contributory factors to self-medication and misuse or overuse of antimicrobials, chief among which are; lack of knowledge about the risk of self-medication, low socioeconomic status, widespread availability of antimicrobials in patent and propriety medicine vendor stores (PPMVs), and other drug retail outlets, and high cost of healthcare delivery in resource-poor setting[2, 5].

Additionally, with the increased global prevalence of antimicrobial self-medication as well as inappropriate use of antimicrobial agents, the burden of antimicrobial drug resistance is likely to persist if nothing is done to stem the tide. This ugly trend has been projected to cause the occurrence of over 10 million deaths from antimicrobial resistance globally by the year 2050[1].

The present review, stands to provide the platform to stay a step ahead by providing a targeted multisectoral approach that can help to solve this hydra-headed problem as well as stem the tide of inappropriate antimicrobial drug use in Nigeria, while still relying heavily on the development of novel antimicrobial agents by pharmaceutical companies

2. MATERIALS AND METHODS

The study entails a systematic review of 18 research articles on antimicrobial self-medication using search engines such as Google Scholar, PubMed, Medline, Hinari, Ebscohost, Embase, and Scopus. Medical subject headings (MSH) such as 'antimicrobials and self-medication' were used for the search. The initial search generated 34 articles which were later pruned to 18 articles using specific inclusion and exclusion criteria. Inclusion criteria used include articles on antimicrobial self-medication in Nigeria between 2000-2021, articles written in the English language, and articles with a focus on factors associated with antimicrobial resistance in Nigeria. All articles outside the scope of the study and which were not written in the English language were excluded from the study. A flow chart diagram (Figure 1) was used in the schematic analysis of the systematic review as suggested by the Prisma statement for conducting systematic reviews and meta-analysis as indicated on the website http://prisma-statement.org/prismastatement/Checklist.aspx.

Analysis was done using proportions, tables, and charts.

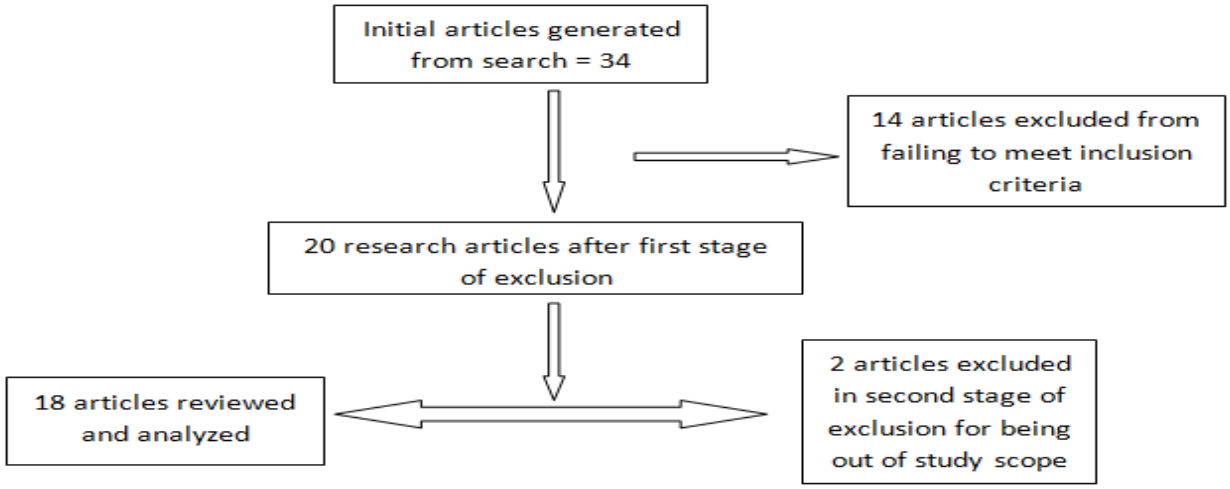


Figure 1: Schematic flow diagram showing article search

3. RESULTS

Of the 18 articles analyzed (consisting of 8,012 participants, of which 3,892 (48.6%) were undergraduate students in various institutions across the country), 10 (55.6%) of the studies were from Southern Nigeria while 8 (44.4%) were from Northern Nigeria. Based on the review, the overall prevalence of ASM in Nigeria was 51.4%.

The majority of the report 17 (94.4%) showed a preponderance of antibiotic self-medication while 1 study (5.6%) focused on anti-malarial self-medication among postpartum mothers (Table 1). Antimicrobials commonly involved in self-medication practice in the population include amoxicillin+clavulanic acid (66.7%), ampicillin+cloxacillin (55.6%), metronidazole (27.8%), ciprofloxacin (27.8%), and tetracycline (16.7%). Perceived medical conditions commonly implicated in the practice of antimicrobial self-medication in Nigeria include fever (66.7%), diarrhea (38.9%), ENT infections (33.3%), and malaria (22.2%).

Table 1: Summary of reviewed articles

Research article	Study type	Key findings
Omolase *et al.*[9]	Cross-sectional (N=200)	There was 85% response rate. Drugs commonly implicated in self-medication include antimalarials (15.9%), antimalarial+analgesics (22.4%), antimalarial+antibiotics (15.3%), and antibiotics +analgesics (10%). Reasons for self-medication includes perception of symptoms as minor (54.7%), and financial constrain (22.4%).
Tobin *et al.*[10]	Cross-sectional (N=800)	The prevalence of ASM was 374 (46.8%). Antimicrobials implicated include ampicillin + cloxacillin (31.3%), and amoxicillin (24.8%). Reasons for practicing ASM include easy access to drugs in drug retail outlets. The medical condition commonly implicated includes RTI's (27.2%). Being elderly and being widowed and divorced were significantly associated with the practice of ASM.
Sapkota *et al.*[11]	Cross-sectional (N=708)	The response rate was 95.4%. 24% reported the practice of ASM (95% CI: 21-27%). Factors associated with the practice of ASM were a low level of education, heavy menstrual bleeding, and perceived non-severe illnesses. Ampicillin, tetracycline, ciprofloxacin, and metronidazole were the most frequently used antimicrobials. Antimicrobials for self-medication were most commonly obtained from the chemist and pharmacy
Khalid *et al.*[12]	Cross-sectional (N=217)	The response rate was 100%. The prevalence of ASM was 92.2%. Reasons for practicing ASM include previous knowledge about medications (40.4%) and no time to see the doctor (27.5%). Drugs commonly implicated in ASM include amoxicillin(32.6%), amoxicillin + clavulanic acid (21.7%), and ciprofloxacin (22.6%). Medical conditions commonly treated include RTIs, diarrhea disease, typhoid, and wound infection. PPMV's and community pharmacy shops were the most common sources of antimicrobials.
Ajibola *et al.*[13]	Cross-sectional (N=1,230)	The response rate was 84.8%. The prevalence of ASM was 43% among undergraduate students and 26% among community members. Local chemist stores and pharmacies (48.4%) were the main sources of medication. Antimicrobials commonly used include metronidazole(18%), amoxicillin + clavulanic acid (16.8%), ampicillin +cloxacillin (14.8%), cotrimoxazole (12%), and tetracycline (11%). Reasons for practicing self-medication include long delays in the hospital (46%, 35%), cheaper medications (26%, 19%), and distance to the hospital (15%, 21%)

Iribhogbe and Odoya [5]	Cross-sectional (N=150)	The prevalence of antimalarial self-medication among postpartum mothers for themselves and their newborns was 42.7% and 22.7% respectively. Reasons adduced for such practice is that malaria is expensive to treat (24.7%). Artemether + lumefantrine combination were the most commonly used drug for self-medication
Esan *et al.*[14]	Cross-sectional (N=384)	The prevalence of antibiotic self-medication was 10.5% while that of antimalarial self-medication was 33%. Reasons for practicing ASM were the unfriendly attitude of health workers (27.7%), lack of time to go to the clinic (26.7%), and far distance of clinic/hospital from the place of residence (15.3%)

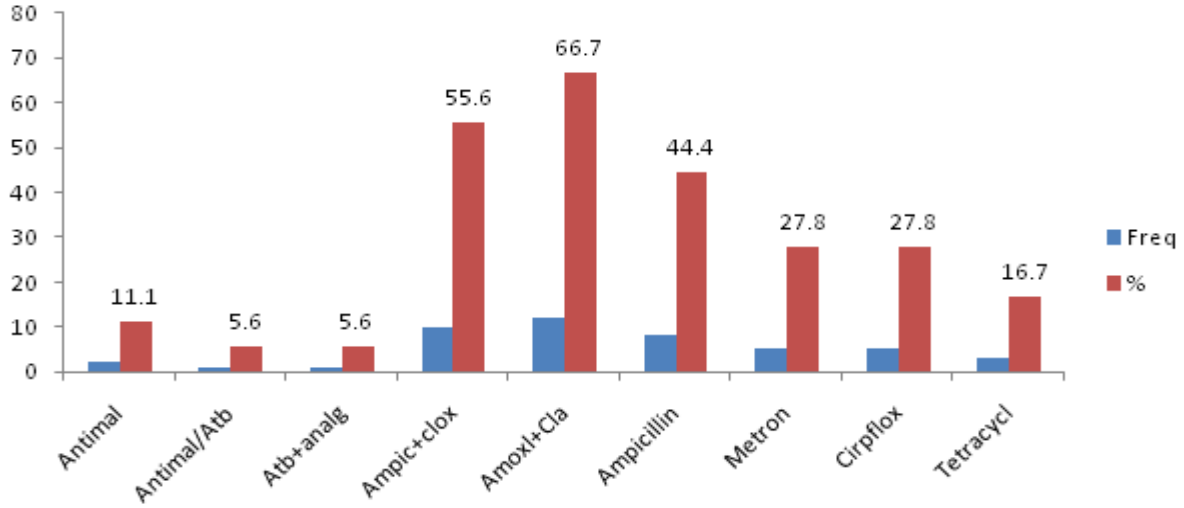


Figure 2: Drugs commonly implicated in self-medication practice in Nigeria
Keys: Antimal=antimalarials, Antimal/Atb=antimalarial+antibiotic, Atb+analg=antibiotic+analgesic, Ampic+clox=ampicillin+cloxacillin, Amoxl+Cla=Amoxicillin+clavulanic acid, Metron=metronidazole, Cirpflox=ciprofloxacin.

4. DISCUSSION

The practice of ASM is a common occurrence in low-and-middle-income countries. Analysis of data from over 31,000 participants from previous studies has shown a prevalence of 38.8% (95% CI: 29.5-48.1%)[2]. Similarly, findings from the present report have shown an overall prevalence of 51.4% in Nigeria which is in tandem with other reports emerging from Nigeria and other sub-Saharan countries[2,5,10,15]. Additionally, the high preponderance of non-prescription drug use of antimicrobials as shown by other systematic reviews is associated with the practice of ASM and other risk factors such as non-compliance with appropriate dosing schedule, inappropriate drug use, sharing of previous prescriptions, and use of left-over drugs[2,5,16,17]. This is consistent with reports from the present review which shows that self-medication was associated with the use of left-over drugs, old prescriptions, prescriptions from friends and family members, and easy access to antimicrobials from drug retail outlets and pharmacy shops. These findings suggest that in Nigeria, antimicrobials which are prescription-only medications are sold as non-prescription drugs due to the lack of regulation of its sales making the drugs readily available for public consumption.

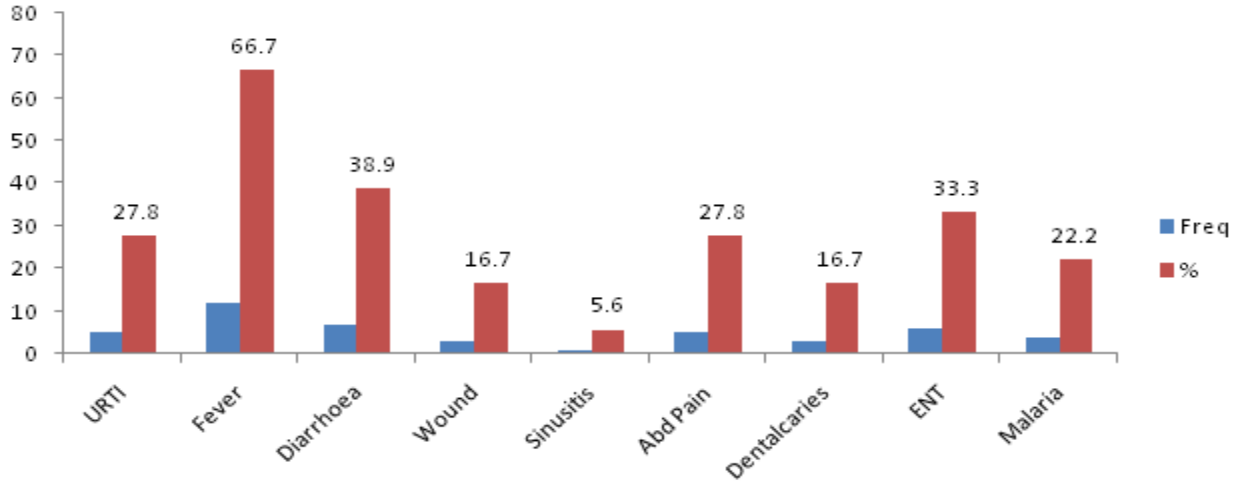


Figure 3: Perceived medical conditions commonly treated during the practice of antimicrobial self-medication

Keys: URTI=upper respiratory tract infections, Abd=abdominal, ENT=ear, nose and throat.

The majority of the studies have reported respiratory tract infections, gastrointestinal symptoms, ENT infections, and malaria as the various illnesses associated with the practice of ASM[9,10,12-13]. Additionally, the present finding is also in keeping with available reports that show that antimicrobials and antimalaria agents were the most common agents involved in self-medication practice in the population. Antibacterial agents such as ampicillin, amoxicillin + clavulanic acid, tetracycline, penicillin, metronidazole, ciprofloxacin, kanamycin, and ceftriaxone were commonly reported while antimalarials such as chloroquine, sulfadoxine + pyrimethamine, artemether + lumefantrine, and quinine were agents mostly implicated in self-medication practice in most low-and-middle-income countries[2,5,18].

From the available reports, the practice of ASM has also been attributed to the perception of illnesses as being mild to moderate, high cost of healthcare delivery, negative attitude of healthcare workers, lack of time to access the clinic/hospital, the distance of clinic/hospital from the area of residence and lack of adequate knowledge about the risk of self-medication. The majority of the studies also reported drug retail outlets, pharmacy shops, left-over drugs, gifts from friends and loved ones, and clinic/hospitals workers as their main sources of drug supply[5,10,13,14,19-20].

The practice of ASM has several implications for the healthcare system. The commonly reported negative outcome of self-medication includes increased incidence of adverse drug reactions, lack of resolution of ailments, delay in recovery from illnesses, and increased mortality from infectious diseases[6,21-23]. Moreover, the poor healthcare system in LMICs has been identified as one of the key factors promoting ASM in sub-Saharan Africa.[2] This is further complicated by the lack of regulatory supervision of over-the-counter sales of prescription-only drugs as is obtainable in other developed countries of the world. Furthermore, striking a balance between individuals who truly need the drugs due to their infectious disease status and those who do not have the infectious disease is a major challenge that requires attention in Nigeria; this is due to the high burden of infectious diseases in resource-poor settings[2,5,6,10,13]. Moreover, the practice of ASM is influenced by the fact that in LMICs patients can purchase drugs without being regulated[2].

Also, antimicrobial misuse can also create problems such as skin reactions, severe allergies, and other forms of hypersensitivity reactions[6,21-23]. Moreover, health care expenditure involving the treatment of antimicrobial drug resistance has increased in most countries including Nigeria due to the practice of ASM[5].

Studies have shown that in a vast majority of cases the emergence of antimicrobial drug resistance stems from irrational drug use, misuse, and overuse of antimicrobial medications with the practice of ASM being the most prevalent cause[8,23-25]. This, coupled with ignorance, lack of knowledge about the appropriate course of antimicrobials and their dosage limit are the major contributory factors to irrational drug use and the incorrect treatment of medical conditions[2,3,7-8,15]. Studies have also shown that patients who perceive their illness as being mild to moderate are less likely to visit healthcare professionals for appropriate treatment of their medical condition[6].

The approach towards the prevention and control of ASM should be multisectoral. Steps should be taken to cut across all levels of the society both at the individual, community, governmental, non-governmental, healthcare professionals, and pharmaceutical industries[2,8]. At the individual level of prevention, the use of antimicrobials as prescription-only medications should be encouraged. Also, all efforts should be geared towards discouraging the practice of ASM through public enlightenment campaigns. The campaigns should capture the discouragement of the use of left-over antimicrobials, the encouragement of compliance with prescribed medications, and the promotion of personal and environmental hygiene in communities. Community leaders should work in conjunction with governmental and non-governmental agencies to ensure the successful implementation of appropriate drug regulation policies that will promote the use of prescription-only medications[5,8]. Further more, a robust platform should be created to strengthen existing regulatory policies and preventive programmes. Also, a platform should be created to foster adequate information dissemination about the dangers of inappropriate antimicrobial utilization[8].

Healthcare professionals should ensure that they prescribe antimicrobials only when needed within the context of strict regulatory guidelines. They should also play a pivotal role in educating patients on the correct use of antimicrobials. Patients should also be educated about disease prevention through vaccination and good personal and environmental hygiene. Healthcare industries in Nigeria should continue to engage in research for the development of novel antimicrobials, vaccines, and new diagnostic techniques[5,8,25].

This multisectoral approach is embedded in the WHO global action plan which focuses on several strategic areas, one of which is to improve the awareness and understanding about antimicrobials, as well as strengthen the optimizing of the use of antimicrobial agents in the population[8]. This approach is being driven by several key initiatives such as world antimicrobial awareness week, global antimicrobial resistance surveillance system, global antimicrobial research, and development partnership as well as interagency coordination groups on antimicrobial drug use and resistance[8,26].

5. CONCLUSION

The systematic review has shown a high prevalence of ASM as well as highlighted key determinants of its practice in Nigeria. A multisectoral approach has been recommended as a reliable public health intervention strategy that can help in mitigating the practice of ASM and its attendant risk to the population both in Nigeria and other regions of the world.

Conflict of Interest Declaration: The author hereby declares that there is no conflict of interest in the conduct of this review.

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